

THE p -ADIC K -THEORY OF SMOOTH ALGEBRAS OVER PERFECTOID RINGS

ABSTRACT. Let A be a perfectoid ring and let R be a smooth A -algebra. We prove that the p -completed algebraic K -theory of R agrees with its p -completed homotopy K -theory. If R/pR has relative dimension at most d over A/pA , we also prove that the fiber of $K(R; \mathbf{F}_p) \rightarrow K(R[1/p]; \mathbf{F}_p)$ has no homotopy groups above degree d . This answers Question 5.14 of Antieau–Mathew–Morrow. The proof constructs a perfectoid coefficient extension which induces injections on the homotopy groups of K -theory modulo p with support on p and receives a map from a mixed-characteristic rank-one perfectoid valuation ring.

1. INTRODUCTION

Antieau, Mathew, and Morrow proved homotopy invariance and a localization bound for the p -adic K -theory of smooth algebras over a perfectoid ring that is an algebra over a perfectoid valuation ring [AMM22, Theorems 5.10 and 5.12]. Their Question 5.14 asks whether these conclusions hold over every perfectoid ring. We prove that they do.

Fix a prime p . We use nonconnective algebraic K -theory. For a spectrum E , write $E/p = \text{cofib}(p : E \rightarrow E)$ and $E_p^\wedge = \varprojlim_n E/p^n$. Accordingly, $K(-; \mathbf{F}_p) = K(-)/p$ and $K(-; \mathbf{Z}_p) = K(-)_p^\wedge$; the same conventions apply to homotopy K -theory KH . A map of spectra is d -truncated if its fiber has vanishing homotopy groups in all degrees greater than d .

Theorem 1.1. *Let A be a perfectoid ring and let R be a smooth A -algebra.*

- (i) *If $A/pA \rightarrow R/pR$ has relative dimension at most d , then*

$$K(R; \mathbf{F}_p) \longrightarrow K(R[1/p]; \mathbf{F}_p)$$

is d -truncated.

- (ii) *The canonical map*

$$K(R; \mathbf{Z}_p) \longrightarrow KH(R; \mathbf{Z}_p)$$

is an equivalence.

In part (i), the map on homotopy groups is therefore an isomorphism in degrees $i \geq d + 2$ and injective in degree $d + 1$. The dimension in the statement is that of the special fiber.

For an ordinary ring T , let $\text{Perf}_p(T)$ be the full subcategory of perfect complexes M such that $M[1/p] = 0$, and set

$$(1.1) \quad F(T) = K(\text{Perf}_p(T); \mathbf{F}_p) \simeq \text{fib}(K(T; \mathbf{F}_p) \longrightarrow K(T[1/p]; \mathbf{F}_p)).$$

The reduction to the known theorems uses the fixed perfect field

$$(1.2) \quad \kappa = \mathbf{F}_p(X_1, X_2, \dots)^{\text{perf}}.$$

For a perfectoid ring A , define

$$(1.3) \quad A_\kappa = (A \otimes_{\mathbf{Z}_p} W(\kappa))_p^{\wedge, L}.$$

Here the superscript denotes derived p -completion. The following proposition gives both the nature of this extension and its role in K -theory.

Proposition 1.2. *Let A be a perfectoid ring.*

- (i) *The derived completion A_κ in (1.3) is an ordinary perfectoid ring.*

- (ii) For every flat A -algebra R , put $R_\kappa = R \otimes_A A_\kappa$. The base-change map induces injections
- $$(1.4) \quad \pi_i F(R) \hookrightarrow \pi_i F(R_\kappa) \quad (i \in \mathbf{Z}).$$
- (iii) If $A \neq 0$, there is a mixed-characteristic rank-one perfectoid valuation ring $\mathcal{O}$ and a unital homomorphism $\mathcal{O} \rightarrow A_\kappa$.

The two substantive parts of the reduction concern the same coefficient extension. To obtain (1.4), we approximate $W(\kappa)$ by polynomial rings with monic polynomials inverted. Localization and additivity give a retraction for one variable and finitely many monic denominators. Iteration and filtered colimits give injections on homotopy groups. Derived reduction modulo p then identifies the supported perfect categories before and after completion.

The valuation ring in part (iii) comes from a different feature of the extension. Write $A_\kappa = W(B_\kappa)/(\xi)$ with B_κ perfect. Multiplying ξ by a suitable Witt-vector unit makes its higher Teichmüller coefficients generate a perfect subfield of B_κ . Its constant coefficient supplies a valuation parameter. An Eisenstein tower over the Witt vectors of that subfield produces $\mathcal{O}$.

Sections 2 and 3 prove parts (i) and (ii) of Proposition 1.2; Section 4 proves part (iii). Theorem 1.1 then follows in Section 5 from the results of [AMM22, Theorems 5.10 and 5.12]. The localization bound passes back along (1.4). For homotopy invariance, we apply the same injection to $R[T]$ and use the splitting of evaluation at $T = 0$.

Conventions. All rings are commutative and unital. Perfectoid rings are integral perfectoid rings in the sense of [CS24, §2.1.2]. For an ordinary ring T and $u \in T$, the notation $T//u$ denotes the derived quotient by u , whose underlying complex is $\text{cofib}(u : T \rightarrow T)$. Thus

$$T_p^{\wedge, L} = \varprojlim_n T//p^n.$$

An undecorated completion denotes ordinary adic completion. We write $T[u^n]$ for the kernel of multiplication by u^n . The notation $\text{Sp}_{\leq d}$ denotes spectra with homotopy groups vanishing above degree d . We use localization, additivity, Morita invariance, and compatibility with filtered colimits for nonconnective K -theory in the setting of small stable categories [BGT13, §§8–9, especially Theorem 9.8].

For a perfect $\mathbf{F}_p$ -algebra B , $W(B)$ denotes the ring of p -typical Witt vectors, and $[b]$ denotes the Teichmüller lift of $b \in B$. Every Witt vector has a unique expansion $\sum_{n \geq 0} p^n [b_n]$. We refer to the b_n as its Teichmüller coefficients.

2. MONIC LOCALIZATION AND COEFFICIENT APPROXIMATION

Lemma 2.1. *Let T be an ordinary ring and let $f(Y) \in T[Y]$ be monic. The map*

$$F(T) \longrightarrow F(T[Y, 1/f])$$

induced by the inclusion of constants has a retraction in spectra.

Proof. Let

$$\mathcal{D} = \text{Perf}(T[Y] \text{ on } (p, f)),$$

where support on (p, f) means that the complex vanishes after inverting either p or f . Localization for nonconnective K -theory gives a cofiber sequence

$$(2.1) \quad K(\mathcal{D}; \mathbf{F}_p) \longrightarrow F(T[Y]) \longrightarrow F(T[Y, 1/f]).$$

Restriction of scalars defines an exact functor

$$U : \mathcal{D} \longrightarrow \text{Perf}_p(T).$$

Indeed, $\mathcal{D}$ is thickly generated by the tensor product of the Koszul complexes on p and f [Stacks, Tags 0A9A and 0GEG]. Since f is monic, it is a nonzerodivisor and $T[Y]/f$ is finite free over T .

The restriction of this generator is consequently a perfect T -complex with support on p . The same holds for its thick closure.

The two-term free resolution of $T = T[Y]/Y$ gives, naturally in $M \in \mathcal{D}$,

$$M \otimes_{T[Y]}^{\mathbf{L}} T \simeq \operatorname{cofib}(Y : U(M) \longrightarrow U(M)).$$

Additivity therefore supplies a nullhomotopy of the composite

$$K(\mathcal{D}; \mathbf{F}_p) \longrightarrow F(T[Y]) \xrightarrow{Y=0} F(T).$$

By (2.1), evaluation factors through a map $F(T[Y, 1/f]) \rightarrow F(T)$. Its composite with the constants map is homotopic to the identity, as is the composite before localization. $\square$

The retraction in Lemma 2.1 is constructed on K -theory spectra. Its usefulness is that the base ring can be recovered on homotopy groups even when evaluation at zero does not extend to the localized ring.

Corollary 2.2. *Suppose T' is obtained from T by finitely many operations of the following form: adjoin one polynomial variable, then invert any set of monic polynomials in that variable. Then $\pi_i F(T) \rightarrow \pi_i F(T')$ is injective for every i .*

Proof. A finite product of monic polynomials is monic, so Lemma 2.1 handles each finite set of denominators. Both F and homotopy groups commute with filtered colimits. It follows that adjoining one variable and inverting any set of monic polynomials preserves the classes from the original ring. Iterating proves the assertion. $\square$

We now construct a single coefficient ring to which this corollary applies and whose completion is $W(\kappa)$.

Proposition 2.3. *There is a flat $\mathbf{Z}_p$ -algebra S and a homomorphism $S \rightarrow W(\kappa)$ which is an equivalence after derived reduction modulo p , such that, for every $\mathbf{Z}_p$ -algebra T ,*

$$(2.2) \quad \pi_i F(T) \longrightarrow \pi_i F(T \otimes_{\mathbf{Z}_p} S)$$

is injective for all $i \in \mathbf{Z}$. Moreover, $S_p^\wedge \cong W(\kappa)$.

Proof. For $m, n \geq 0$, construct $S_{m,n}$ from $\mathbf{Z}_p$ by adjoining independent variables $Y_{1,n}, \dots, Y_{m,n}$ in that order. After each adjunction, invert every monic polynomial in the new variable over the preceding ring. Each $S_{m,n}$ is p -torsion-free, and hence flat over $\mathbf{Z}_p$. Induction on m gives

$$(2.3) \quad S_{m,n}/p \cong \mathbf{F}_p(Y_{1,n}, \dots, Y_{m,n}).$$

For this induction, every monic polynomial over the preceding residue field lifts to one over the preceding ring, and every nonzero polynomial over a field is a unit times a monic polynomial.

There are transition maps increasing m , and maps

$$S_{m,n} \longrightarrow S_{m,n+1}, \quad Y_{i,n} \longmapsto Y_{i,n+1}^p.$$

The latter extend over the localizations: substitution preserves monicity in the variable being adjoined, with the coefficients mapped through the preceding stages. Set

$$S = \varinjlim_{m,n} S_{m,n}.$$

It is p -torsion-free and (2.3) identifies S/p with κ .

The assignments

$$Y_{i,n} \longmapsto [X_i^{1/p^n}]$$

define compatible maps $S_{m,n} \rightarrow W(\kappa)$. At each stage an inverted polynomial has nonzero reduction in κ , since its new variable is transcendental over the preceding residue field. Its

image in $W(\kappa)$ is therefore a unit. The resulting map $S \rightarrow W(\kappa)$ induces the given identification modulo p . Since both rings are p -torsion-free, it is a derived mod- p equivalence. Induction using multiplication by p identifies their quotients modulo p^r for every r , and hence their ordinary completions.

After base change to any $T/\mathbf{Z}_p$, the construction of each $T \otimes_{\mathbf{Z}_p} S_{m,n}$ has the form in Corollary 2.2. Thus the map from $\pi_i F(T)$ to each finite stage is injective. If a class from $\pi_i F(T)$ vanishes in the filtered colimit, it vanishes at one such stage. This proves (2.2). $\square$

3. THE COMPLETED COEFFICIENT EXTENSION

We prove parts (i) and (ii) of Proposition 1.2. The point of the completion argument is to identify the derived reduction modulo p of the specified coefficient map. The following lemma then identifies the supported perfect categories.

Lemma 3.1. *Let $T \rightarrow T'$ be a map of ordinary rings such that $T//p \rightarrow T'//p$ is an equivalence. Base change induces an equivalence*

$$\mathrm{Perf}_p(T) \xrightarrow{\sim} \mathrm{Perf}_p(T').$$

Proof. The Koszul complex $P = \mathrm{cofib}(p : T \rightarrow T)$ thickly generates $\mathrm{Perf}_p(T)$, and its base change P' thickly generates $\mathrm{Perf}_p(T')$. By hypothesis, $P \rightarrow P'$ is an equivalence after restriction to T . Adjunction therefore identifies the base-change map of endomorphism algebras with an equivalence:

$$\mathrm{End}_T(P) \longrightarrow \mathrm{End}_{T'}(P') \simeq \mathrm{RHom}_T(P, P').$$

Morita theory for the thick subcategories generated by P and P' proves the assertion. $\square$

Lemma 3.2. *Let C be a reduced ring, let $t \in C$, and let $\widehat{C} = \varprojlim_n C/t^n C$. Then the natural map*

$$C//t \longrightarrow \widehat{C}//t$$

is an equivalence. If C is a perfect $\mathbf{F}_p$ -algebra, then $\widehat{C}$ is perfect.

Proof. If $t^n x = 0$, then $(tx)^n = 0$, so reducedness gives $tx = 0$. Thus $C[t^n] = C[t]$ for $n \geq 1$. Also $C[t] \cap tC = 0$: if $x = ty$ and $tx = 0$, then $t^2 y = 0$, whence $ty = 0$. For every $n \geq 1$ there is consequently an exact sequence

$$(3.1) \quad 0 \longrightarrow C[t] \longrightarrow C/t^n C \xrightarrow{t} C/t^{n+1} C \longrightarrow C/tC \longrightarrow 0.$$

The transition maps on the intervening images $tC/t^{n+1}C$ are surjective. Taking inverse limits in the two short exact sequences underlying (3.1) gives

$$0 \longrightarrow C[t] \longrightarrow \widehat{C} \xrightarrow{t} \widehat{C} \longrightarrow C/tC \longrightarrow 0.$$

This identifies both homotopy groups of the derived quotients, compatibly with the natural map, proving the first assertion.

If C is perfect, Frobenius carries $t^n C$ onto $t^{pn} C$. Frobenius and its inverse are therefore continuous for the t -adic topology and extend to mutually inverse endomorphisms of $\widehat{C}$. $\square$

We use the following form of the perfectoid presentation criterion [CS24, (2.1.2.1)]: a ring is perfectoid if it is of the form

$$(3.2) \quad W(B)/(\xi), \quad \xi = [t] + p\eta, \quad \eta \in W(B)^\times,$$

where B is a perfect $\mathbf{F}_p$ -algebra complete for the t -adic topology. For a perfectoid ring A , one can take $B = A^\flat$. The unit condition in (3.2) is equivalent to invertibility of the Witt coordinate of index 1 in the cited criterion.

Lemma 3.3. *Let B be a perfect $\mathbf{F}_p$ -algebra and let $\xi = [t] + p\eta \in W(B)$, where η is a unit. For every perfect B -algebra D , the image of ξ in $W(D)$ is a nonzerodivisor.*

Proof. The reduced ring D embeds into

$$\prod_{\mathfrak{q} \in \operatorname{Spec} D} \operatorname{Frac}(D/\mathfrak{q}).$$

Each field in this product is perfect. Witt vectors preserve injections and products. In a field factor where $t \neq 0$, the image of ξ has nonzero reduction modulo p and is a unit. In a factor where $t = 0$, it is p times a unit. In either case multiplication by ξ is injective. The same is therefore true in $W(D)$. $\square$

Proof of Proposition 1.2, parts (i) and (ii). Choose a presentation

$$B = A^b, \quad A = W(B)/(\xi), \quad \xi = [t] + p\eta$$

as in (3.2), and put

$$(3.3) \quad C = B \otimes_{\mathbf{F}_p} \kappa, \quad B_\kappa = \widehat{C}^t.$$

The ring C is perfect: its Frobenius is the tensor product of the Frobenius automorphisms of B and κ . Lemma 3.2 shows that B_κ is perfect and that

$$(3.4) \quad C//t \xrightarrow{\sim} B_\kappa//t.$$

The presentation criterion implies that $W(B_\kappa)/(\xi)$ is an ordinary perfectoid ring. It is also derived p -complete. Indeed, $W(B_\kappa)$ is p -torsion-free and p -adically complete, and Lemma 3.3 identifies its quotient by ξ with the cofiber of multiplication by ξ . Derived p -complete complexes are closed under cofibers.

Let $S \rightarrow W(\kappa)$ be the map from Proposition 2.3. The maps from $W(B)$ and $W(\kappa)$ to $W(B_\kappa)$ give a homomorphism

$$(3.5) \quad A \otimes_{\mathbf{Z}_p} S \longrightarrow W(B_\kappa)/(\xi).$$

We compute its derived reduction modulo p . Since ξ is regular and S is flat over $\mathbf{Z}_p$, the source reduces to

$$(A \otimes_{\mathbf{Z}_p} S)//p \simeq (B//t) \otimes_{\mathbf{F}_p}^{\mathbf{L}} \kappa \simeq C//t.$$

Applying Lemma 3.3 over B_κ identifies the target's derived reduction with $B_\kappa//t$. Under these identifications, the map is (3.4). Thus (3.5) is a derived mod- p equivalence.

Taking derived p -completions gives

$$(A \otimes_{\mathbf{Z}_p} S)_p^{\wedge, L} \simeq W(B_\kappa)/(\xi).$$

The map $S \rightarrow W(\kappa)$ is itself a derived mod- p equivalence. Both coefficient rings are flat over $\mathbf{Z}_p$, so their derived tensor products with A are ordinary tensor products. Consequently

$$(3.6) \quad A_\kappa \simeq W(B_\kappa)/(\xi).$$

This proves part (i).

For a flat A -algebra R , tensor (3.5) derivedly with R over A . Flatness identifies the resulting source and target with the ordinary rings

$$R \otimes_{\mathbf{Z}_p} S \longrightarrow R \otimes_A A_\kappa = R_\kappa.$$

This is again a derived mod- p equivalence. Lemma 3.1 therefore gives

$$F(R \otimes_{\mathbf{Z}_p} S) \simeq F(R_\kappa).$$

Together with (2.2), this proves (1.4) and part (ii). $\square$

4. A PERFECTOID VALUATION RING MAPPING TO THE EXTENDED BASE

We prove part (iii) of Proposition 1.2. Retain the presentation and notation of (3.3) and (3.6), and assume $A \neq 0$. We will first change the generator ξ so that its higher Teichmüller coefficients lie in a perfect subfield of B_κ . We then realize its constant coefficient as the image of a valuation parameter. This produces the homomorphism of untilts.

4.1. A field of Teichmüller coefficients. As a B -module, $C = B \otimes_{\mathbf{F}_p} \kappa$ is a direct sum of copies of B . Since B is t -adically separated, so is C , and C embeds in B_κ . In particular, $B_\kappa \neq 0$. Let

$$(4.1) \quad J = \sqrt{tB_\kappa}.$$

Completeness implies $t \in \text{Jac}(B_\kappa)$, by the geometric series for $1 - tb$. The Jacobson radical is a radical ideal, so $J \subseteq \text{Jac}(B_\kappa)$. The quotient B_κ/J is thus nonzero, and the unital map $\kappa \rightarrow B_\kappa/J$ is injective. Moreover, this quotient is perfect: Frobenius is surjective because B_κ is perfect and injective because J is radical.

In $W(B_\kappa)$, set

$$w = 1 + \sum_{r \geq 1} p^r [X_r], \quad v = \eta^{-1}w,$$

and write

$$\xi v = \sum_{n \geq 0} p^n [c_n].$$

In $W(B_\kappa/J)$, the equality $[t] = 0$ gives $\xi v = pw$. Uniqueness of the Teichmüller expansion over the perfect quotient yields

$$(4.2) \quad c_1 \equiv 1 \pmod{J}, \quad c_{r+1} \equiv X_r \pmod{J} \quad (r \geq 1).$$

Thus c_1 is a unit. Normalize the generator once more:

$$(4.3) \quad \xi' = \xi v [c_1^{-1}] = \sum_{n \geq 0} p^n [a_n].$$

Multiplication by a Teichmüller lift scales each coefficient of this expansion, so, with $\eta_0 = \eta \bmod p$,

$$(4.4) \quad a_1 = 1, \quad a_0 = t\eta_0^{-1}c_1^{-1}, \quad a_{r+1} \equiv X_r \pmod{J} \quad (r \geq 1).$$

In particular, $(a_0) = (t)$ and $(\xi') = (\xi)$.

For every nonzero polynomial $P \in \mathbf{F}_p[Z_1, \dots, Z_m]$, the residue of $P(a_2, \dots, a_{m+1})$ modulo J is $P(X_1, \dots, X_m)$. This is a nonzero element of the field κ , and hence a unit in B_κ/J . Since $J \subseteq \text{Jac}(B_\kappa)$, the element $P(a_2, \dots, a_{m+1})$ is a unit in B_κ . Thus evaluation at $a_2, a_3, \dots$ embeds the rational function field into B_κ . Perfectness of B_κ extends this embedding uniquely to its perfection. We obtain a perfect subfield

$$(4.5) \quad k' = \mathbf{F}_p(a_2, a_3, \dots)^{\text{perf}} \subset B_\kappa.$$

The normalized relation now has the form

$$(4.6) \quad \xi' = [a_0] + pU, \quad U = 1 + \sum_{r \geq 1} p^r [a_{r+1}] \in W(k')^\times.$$

4.2. The valuation parameter and its untill. Introduce an indeterminate z over k' , and form

$$(4.7) \quad V_0 = \bigcup_{n \geq 0} k'[z^{1/p^n}]_{(z^{1/p^n})}, \quad V = \widehat{V_0}^z.$$

Every nonzero element of V_0 is a unit times z^q for some $q \in \mathbf{Z}[1/p]_{\geq 0}$. Hence V_0 is a rank-one valuation ring with value group $\mathbf{Z}[1/p]$. Its z -adic completion V is the valuation ring in the completion of its valued fraction field, with the same value group. Frobenius and its inverse are continuous and extend from V_0 , so V is perfect.

The assignments

$$(4.8) \quad z^{1/p^n} \mapsto a_0^{1/p^n}$$

define compatible homomorphisms from the localizations in (4.7). Indeed, a polynomial with nonzero constant coefficient maps to an element of $k'^{\times} + J$, since every a_0^{1/p^n} lies in J . Such an element is a unit in B_{κ} . The equality $(a_0) = (t)$ makes this map continuous and allows it to extend to

$$V \longrightarrow B_{\kappa}.$$

Consider the ring

$$(4.9) \quad \mathcal{O} = W(V)/([z] + pU),$$

where U is as in (4.6). We identify it with the completion of an explicit tower of discrete valuation rings. For $n \geq 0$, let

$$(4.10) \quad D_n = W(k')[x_n]/(x_n^{p^n} + pU), \quad x_n \mapsto x_{n+1}^p.$$

Since $W(k')$ is a complete discrete valuation ring with uniformizer p , each defining polynomial is Eisenstein. Thus D_n is a complete discrete valuation ring with uniformizer x_n and ramification index p^n . The transition maps are injective and preserve the valuations normalized by $v(p) = 1$. Their union $D_{\infty} = \varinjlim_n D_n$ is a rank-one valuation ring with value group $\mathbf{Z}[1/p]$. Its p -adic completion $\widehat{D}_{\infty}$ is again such a valuation ring.

Modulo p , completion of the p -torsion-free ring D_{∞} does not change the quotient, and therefore

$$(4.11) \quad \widehat{D}_{\infty}/p \cong \varinjlim_n k'[x_n]/(x_n^{p^n}).$$

Frobenius on this ring is surjective: take roots of the coefficients in k' and replace x_n by x_{n+1} . Its kernel is generated by x_1 , since an element has zero p -th power modulo p exactly when its valuation is at least $1/p$. Together with $x_1^p = -pU$, this is the perfectoid criterion of [CS24, §2.1.2, following (2.1.2.3)]. Thus $\widehat{D}_{\infty}$ is a mixed-characteristic perfectoid valuation ring.

The assignments $x_n \mapsto [z^{1/p^n}]$ satisfy the relations in (4.10) in the ring $\mathcal{O}$. To pass to the completion, first note that $\mathcal{O}$ is p -torsion-free. If

$$pb = ([z] + pU)c \quad \text{in } W(V),$$

then reduction modulo p gives $z\bar{c} = 0$. The ring V is a domain, so $c = pc_1$; cancellation of p then gives $b = ([z] + pU)c_1$. Also $[z] + pU$ is regular by Lemma 3.3. As a regular quotient of the derived p -complete ring $W(V)$, the ring $\mathcal{O}$ is derived p -complete. Its p -torsion-freeness implies ordinary p -adic completeness. We therefore have a homomorphism

$$(4.12) \quad \widehat{D}_{\infty} \longrightarrow \mathcal{O}.$$

The reduction of (4.12) modulo p is the isomorphism

$$\varinjlim_n k'[x_n]/(x_n^{p^n}) \xrightarrow{\sim} V/(z).$$

Indeed, completion does not change $V_0/(z)$, and localization at (z^{1/p^n}) does not change $k'[z^{1/p^n}]/(z)$. Both rings in (4.12) are p -torsion-free and p -adically complete. Induction modulo p^r , followed by passage to the inverse limit, shows that (4.12) is an isomorphism. This proves that $\mathcal{O}$ is a mixed-characteristic rank-one perfectoid valuation ring.

Finally, $W(V) \rightarrow W(B_\kappa)$ sends $[z] + pU$ to $[a_0] + pU = \xi'$. By (4.3) and (3.6), it induces

$$(4.13) \quad \mathcal{O} \longrightarrow W(B_\kappa)/(\xi') = W(B_\kappa)/(\xi) \cong A_\kappa.$$

This proves part (iii) of Proposition 1.2.

5. LOCALIZATION AND HOMOTOPY INVARIANCE

Proof of Theorem 1.1. The zero ring gives zero K -theory spectra, so we may assume $A \neq 0$. Form A_κ and $R_\kappa = R \otimes_A A_\kappa$. Proposition 1.2 makes A_κ a perfectoid algebra over a perfectoid valuation ring $\mathcal{O}$. Moreover, R_κ is smooth over A_κ .

The localization bound. There is an identification

$$R_\kappa/pR_\kappa \cong (R/pR) \otimes_{A/pA} (A_\kappa/pA_\kappa).$$

Thus the special fiber still has relative dimension at most d . The hypotheses of [AMM22, Theorem 5.10] apply to $\mathcal{O} \rightarrow A_\kappa \rightarrow R_\kappa$, giving $F(R_\kappa) \in \mathrm{Sp}_{\leq d}$. Since R is flat over A , Proposition 1.2(ii) gives

$$\pi_i F(R) \hookrightarrow \pi_i F(R_\kappa) = 0 \quad (i > d).$$

By (1.1), this proves Theorem 1.1(i).

Polynomial homotopy invariance. Define

$$(5.1) \quad N_p(R) = \mathrm{fib}(K(R[T]; \mathbf{F}_p) \xrightarrow{T=0} K(R; \mathbf{F}_p)).$$

Finite- p -coefficient K -theory is homotopy invariant on rings in which p is invertible [Wei89, Proposition 1.6]; this application is also used in [AMM22, proof of Theorem 5.12]. Comparing the localization fiber sequences for $R[T]$ and R therefore gives a natural equivalence

$$(5.2) \quad N_p(R) \simeq \mathrm{fib}(F(R[T]) \xrightarrow{T=0} F(R)).$$

Evaluation admits the constants map as a section, so

$$(5.3) \quad F(R[T]) \simeq F(R) \oplus N_p(R).$$

This splitting is compatible with base change in R .

Apply [AMM22, Theorem 5.12] to R_κ and $R_\kappa[T]$. Homotopy invariance of KH and reduction modulo p give $N_p(R_\kappa) = 0$. The algebra $R[T]$ is flat over A , so Proposition 1.2(ii) gives an injection

$$\pi_i F(R[T]) \hookrightarrow \pi_i F(R_\kappa[T]).$$

By (5.3), its restriction to $\pi_i N_p(R)$ factors through $\pi_i N_p(R_\kappa) = 0$. Hence $N_p(R) = 0$.

The same argument applies to every polynomial extension $R[T_1, \dots, T_m]$, which remains smooth over A . Consequently the canonical map from $K(R; \mathbf{F}_p)$ to each term of the simplicial polynomial spectrum defining $\mathrm{KH}(R; \mathbf{F}_p)$ is an equivalence. Geometric realization yields

$$K(R; \mathbf{F}_p) \xrightarrow{\sim} \mathrm{KH}(R; \mathbf{F}_p).$$

The coefficient cofiber sequences imply the corresponding equivalences modulo p^n for all $n \geq 1$. Taking their inverse limit proves

$$K(R; \mathbf{Z}_p) \xrightarrow{\sim} \mathrm{KH}(R; \mathbf{Z}_p),$$

as required. □

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